PRIYANSHU SHARMA

Java Backend • AI/ML • Cloud

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EDUCATION

Bal Vidya Bhawan Public School (CISCE) | Hardoi — 2019 - 2020
10th (HighSchool) — *Percentage : 79%*

Bal Vidya Bhawan Public School (CISCE) | Hardoi , India — 2020 - 2022
12th (Intermediate) — *Percentage : 85.8%*

Pranveer Singh Institute of Technology (PSIT) | Kanpur — 2024 - 2028
Bachelor of Technology (B.Tech) — *CGPA : 8*

- **B.Tech Computer Science & Engineering**, expected graduation in 2028

PROJECTS

Geospatial AI/ML Framework for Urban Heat Stress Mitigation | Github — May 2026 - Present
Lead Developer | Based on an ISRO Urban Heat Mitigation Problem Statement

- Built a **multi-city geospatial AI pipeline** across **Delhi, Bengaluru, Lucknow, and Kanpur**, processing **70,595 spatial observations** for urban heat analysis.
- Fused **6 satellite, climate, and terrain sources** including Landsat 8/9, Sentinel-2, ERA5-Land, ESA WorldCover, GHSL, and SRTM DEM.
- Engineered **24+ thermal and urban-morphology features** using OSM/Overture data, including **1.55M Bengaluru building footprints** and 500 m spatial buffers.
- Trained an **XGBoost LST model** achieving **R^2 = 0.8904, MAE = 1.13°C, and RMSE = 1.50°C** across a **70K+ sample** multi-city dataset.
- Performed **spatial block validation and SHAP explainability**, achieving mean spatial **$R^2 \approx 0.792$** while identifying built-up intensity and land use as key predictors.
- Developed hotspot mapping and **counterfactual cooling scenarios** to compare vegetation, built-up, and urban-form interventions through an interactive geospatial dashboard.

Technologies / Tools Used : *Python, XGBoost, SHAP, Google Earth Engine, GeoPandas, OSM, Overture Maps, Landsat 8/9, Sentinel-2, ERA5-Land, FastAPI, React, TypeScript*

Vidyut — Agentic EV Charging Platform • Startup | Github Vidyut v1.1 — Jan 2026 - Present
Founder & Backend/AI Developer

- Built a **Java Spring Boot backend** for EV charging with **3 user roles**, covering charger discovery, booking, trip planning, and marketplace workflows.
- Designed **REST APIs and PostgreSQL models** with layered architecture, DTO validation, JWT authentication, and role-based access across **4+ domains**.
- Developed an **AI trip-planning agent** evaluating **6+ factors**, including battery state, charger availability, travel time, pricing, and safety reserve.
- Validated the planner on a **~601 km route**, generating a **6-stop charging plan at ~₹985** with automatic rerouting after charger failure.
- Designed an EV charging marketplace connecting drivers, charging companies, and property owners with a target **5–6% platform fee** per transaction.

Technologies / Tools Used : *React, TypeScript, Java, SpringBoot, PostgreSQL, REST APIs, JWT, Maven, Gemini, Google ADK*

SHARMAN — Safety-First Multi-Agent Windows Operator | Github — Sep 2026 - Present
Developer | Ongoing

- Migrating SHARMAN toward a **Rust + Tauri desktop architecture** for secure Windows automation and multi-agent workflows.
- Designing shared task and file-state synchronization to prevent agents from acting on outdated workspace context.
- Designing policy-based execution controls with user approval, sandbox isolation, rollback, and post-action verification.
- Building an auditable action pipeline to track agent identity, task state, approvals, and verified system changes.
- Exploring secure integrations for browser, file-system, Git, terminal, and developer-tool automation.

Technologies / Tools Used : *Rust, Tauri, React, TypeScript, PowerShell, Docker, Windows Sandbox*

Text Pro Editor (by Coursera) | Github — Nov 2025 - Jan 2026
Developer

- Built a React-based smart text editor focused on **efficient text processing, custom data structures, and real-time editing features**.
- **Optimized parsing performance** via `EfficientDocument.java` using memoization; achieved linear *O(N)* time and a **1.8x speedup** on 110K+ character files.
- **Built a custom** `MyLinkedList` **architecture** with JUnit; benchmarks against a `HashSet` proved a **99.9% lookup latency drop** (169.9ms to 21.1µs) at 60K words.
- **Deployed a Trie data structure** to facilitate high-speed keyword indexing, rapid text retrieval, and low-latency **predictive word autocomplete**.
- **Implemented distance-based algorithms** to power an intelligent spell-checking engine that accurately detects typos and suggests real-time corrections.

Technologies / Tools Used : *Java, React, CSS, HTML, TypeScript*

SKILLS

Programming Languages :	Java, Python, JavaScript, TypeScript, SQL, C
Backend Development :	Spring Boot, REST APIs, JWT Authentication
Frontend Development :	HTML, CSS, React.js
Databases :	PostgreSQL, MySQL
Cloud & Deployment :	Google Cloud Run, Cloud SQL, Firebase, Docker
AI/ML & Data :	XGBoost, SHAP, Google Earth Engine, GeoPandas, Machine Learning
Data Analytics & BI :	Power BI, Excel, Tableau, DAX, Data Modeling, Data Visualization
Developer Tools :	Git, GitHub, Maven, YAML, Linux, JUnit, IntelliJ IDEA, VS Code

CERTIFICATIONS & TRAINING

- Completed Infosys Springboard training, gaining proficiency in **Python, HTML, CSS, and Data Science**.
- Completed ***Data Structures and Performance using Java*** (University of California) through Coursera.
- Earned ***Java (Basic)*** Certification from **HackerRank.**
- Completed the **JP Morgan Software Engineering Virtual Experience** through Forage.

CODING PLATFORMS

- ***HackerRank:*** 4□ in C, 4□ in Java, 3□ in Python, 3□ in C++, and 4□ Problem Solving
- ***LeetCode:*** Solved **326+ problems**; ranked around **top 50%** as of Sep 2026
- ***Codeforces:*** Actively participating in contests and practicing competitive programming

STRENGTHS

Analytical Thinking • Quick Learning • Adaptability • Team Collaboration

AWARDS & ACHIEVEMENTS

- National Finalist, Codecure-AI Biohackathon — SPIRIT'26, IIT (BHU), Varanasi
- Advanced to Round 2 — Tata Imagination Challenge 2026

INTERESTS

Chess.com ratings: **1200 Rapid**, **850 Blitz**, **800 Bullet** @unexpectedpriyanshu.